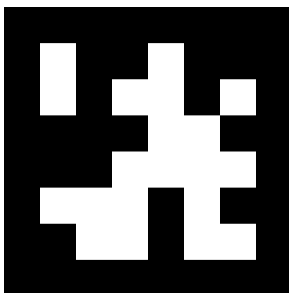


01. Fraction Kit — Fractions

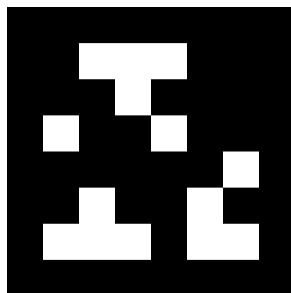
Classes 3–5

LEARNING OBJECTIVE

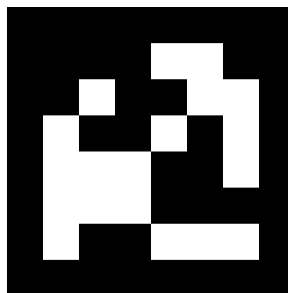
Build and compare fractions using physical pieces.



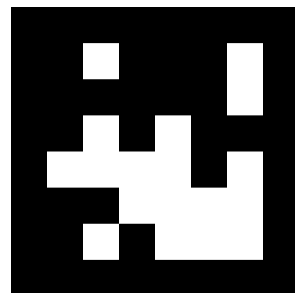
TAG #1
base



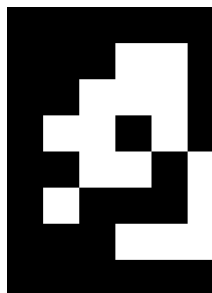
TAG #2
 $\frac{1}{2}$



TAG #3
 $\frac{1}{4}$



TAG #4
 $\frac{1}{8}$



TAG #5
 $\frac{1}{8}$

PHYSICAL TEST

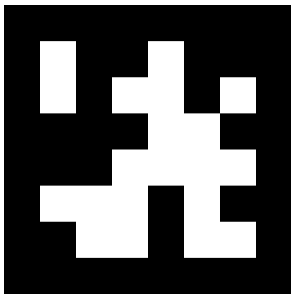
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

02. Decimal Kit — Tenths & Hundredths

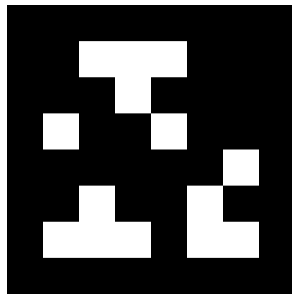
Classes 3–5

LEARNING OBJECTIVE

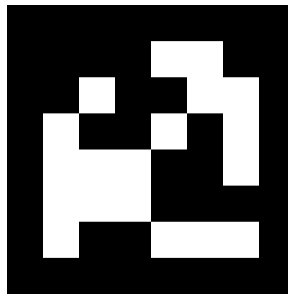
Compose decimals from tenths and hundredths.



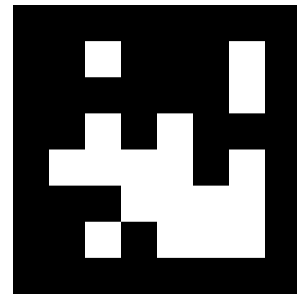
TAG #1
base



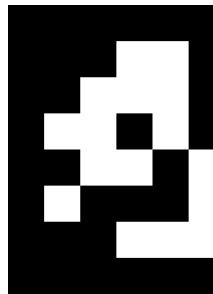
TAG #2
0.10



TAG #3
0.01



TAG #4
0.20



TAG #5
0.02

PHYSICAL TEST

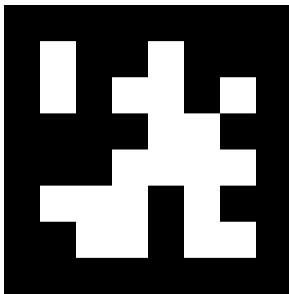
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

03. Fake Money Kit — Counting & Change

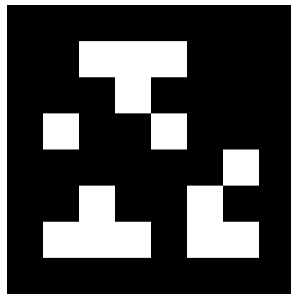
Classes 3–5

LEARNING OBJECTIVE

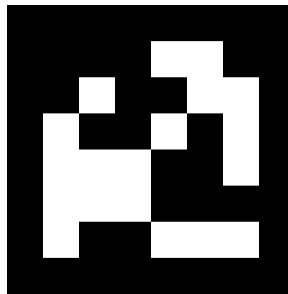
Recognise denominations, add values and practise change.



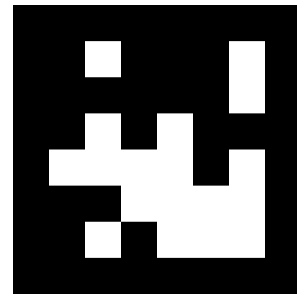
TAG #1
base



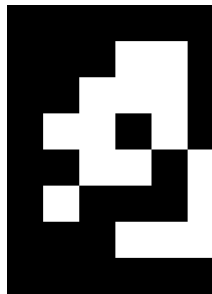
TAG #2
■10



TAG #3
■20



TAG #4
■50



TAG #5
■100

PHYSICAL TEST

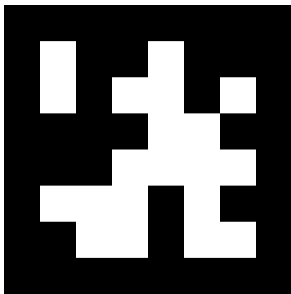
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

04. Rangometry Kit — Angles

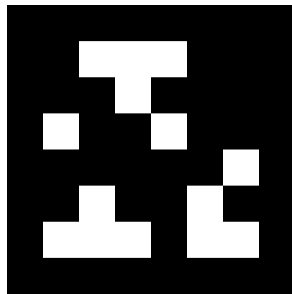
Classes 3–5

LEARNING OBJECTIVE

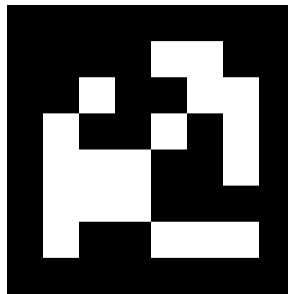
Rotate a physical rod and read the live angle.



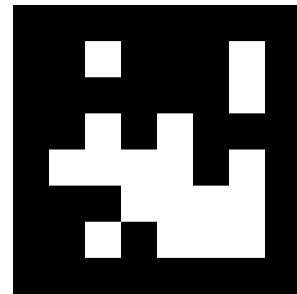
TAG #1
fixed pivot



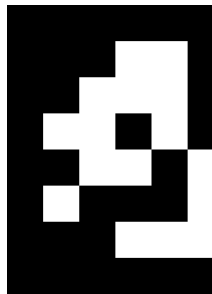
TAG #2
moving rod



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

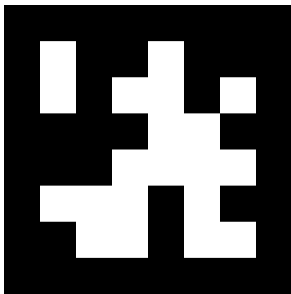
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

05. Training Clock — Time

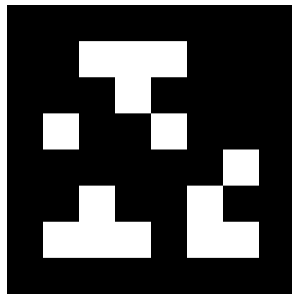
Classes 3–5

LEARNING OBJECTIVE

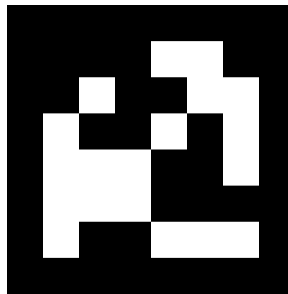
Read analogue time from a physical hand position.



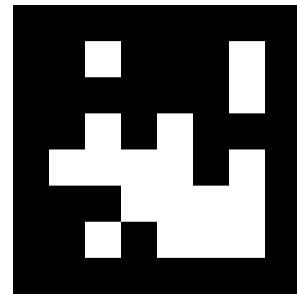
TAG #1
clock centre



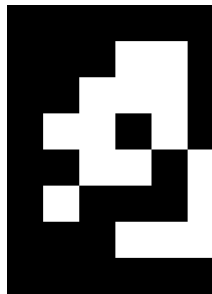
TAG #2
moving hand



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

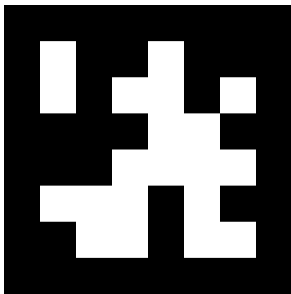
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

06. Sau Rang Ki Khoj — Colour Patterns

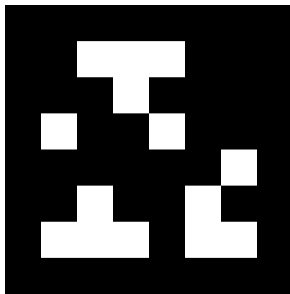
Classes 3–5

LEARNING OBJECTIVE

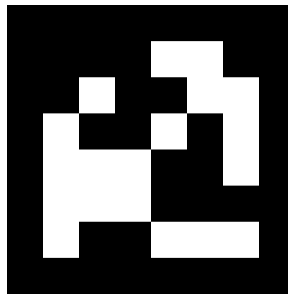
Build repeating patterns with physical colour pieces.



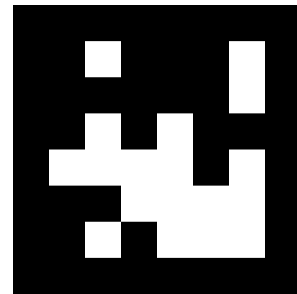
TAG #1
base



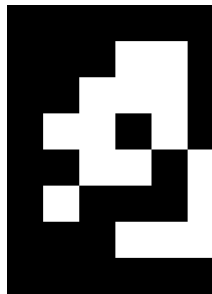
TAG #2
piece A



TAG #3
piece B



TAG #4
piece A



TAG #5
piece B

PHYSICAL TEST

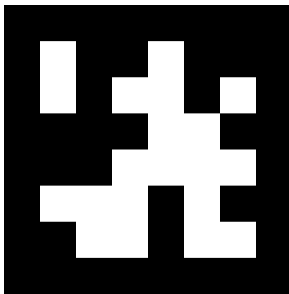
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

07. Tessellation Kit — Shapes & Patterns

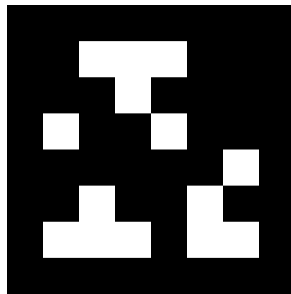
Classes 3–5

LEARNING OBJECTIVE

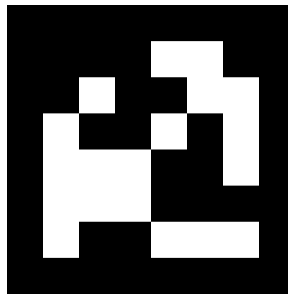
Explore repeating shapes, sides and coverage.



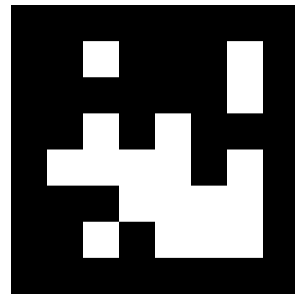
TAG #1
base



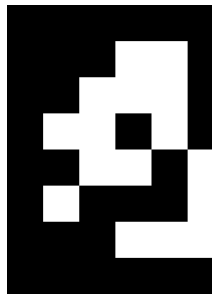
TAG #2
shape



TAG #3
shape



TAG #4
shape



TAG #5
shape

PHYSICAL TEST

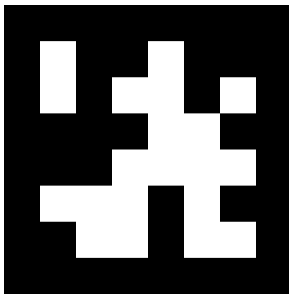
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

08. Dienes Blocks — Place Value

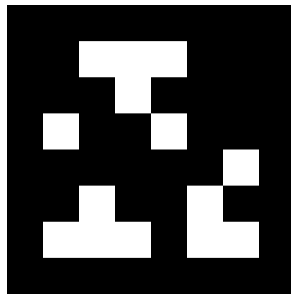
Classes 3–5

LEARNING OBJECTIVE

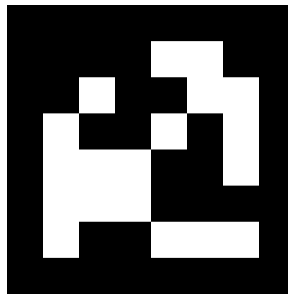
Compose numbers from physical place-value blocks.



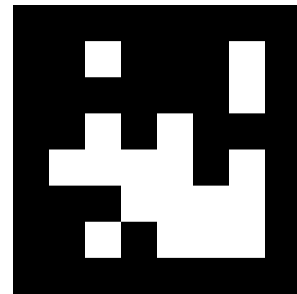
TAG #1
base



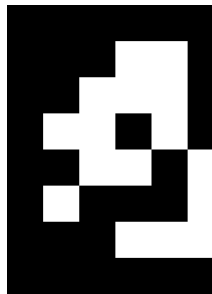
TAG #2
thousands



TAG #3
hundreds



TAG #4
tens



TAG #5
ones

PHYSICAL TEST

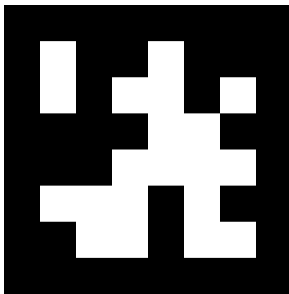
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

09. Place Value / Number Cards

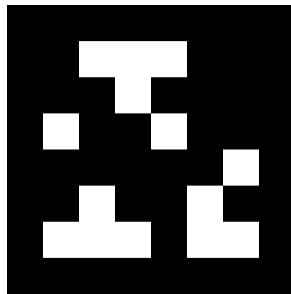
Classes 3–5

LEARNING OBJECTIVE

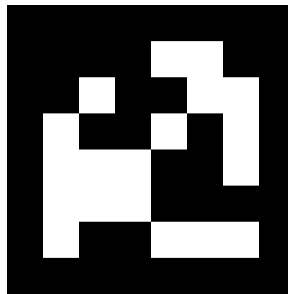
Build and decompose numbers by place value.



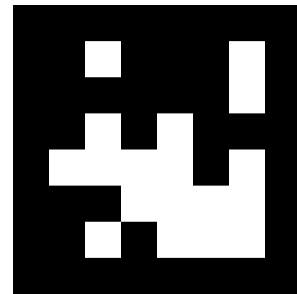
TAG #1
base



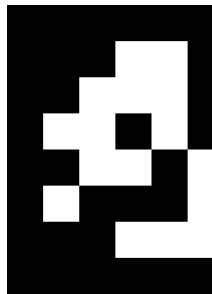
TAG #2
thousands



TAG #3
hundreds



TAG #4
tens



TAG #5
ones

PHYSICAL TEST

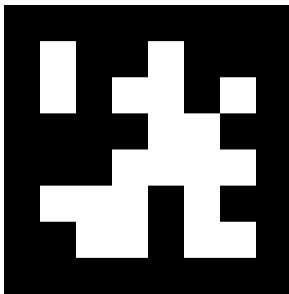
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

10. Probability Kit — Experimental Probability

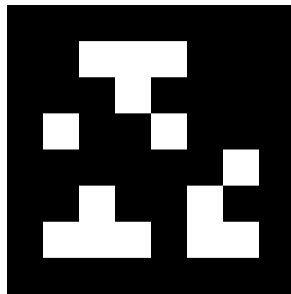
Classes 6–8

LEARNING OBJECTIVE

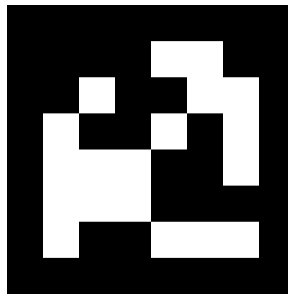
Use physical outcomes to estimate experimental probability.



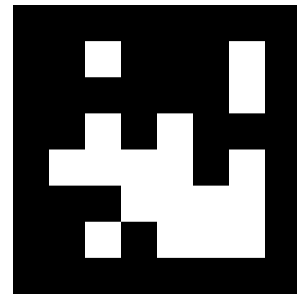
TAG #1
trial board



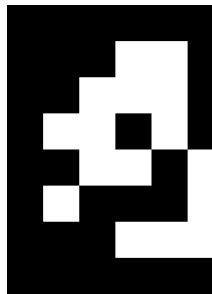
TAG #2
favourable



TAG #3
outcome



TAG #4
outcome



TAG #5
outcome

PHYSICAL TEST

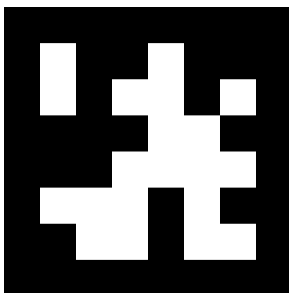
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

11. Fraction Kit — Equivalent Fractions

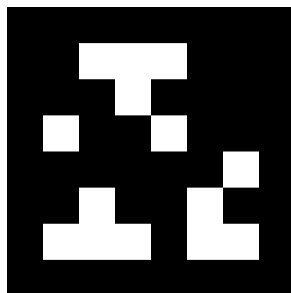
Classes 6–8

LEARNING OBJECTIVE

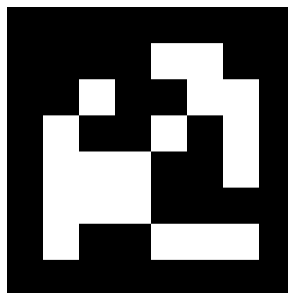
Compare equivalent fraction representations.



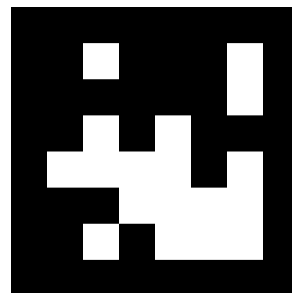
TAG #1
base



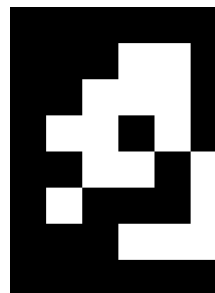
TAG #2
 $\frac{1}{2}$



TAG #3
 $\frac{2}{4}$



TAG #4
 $\frac{4}{8}$



TAG #5
comparison

PHYSICAL TEST

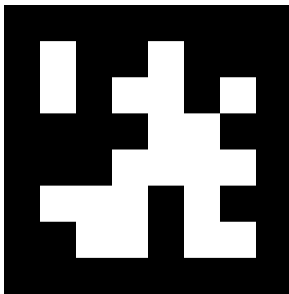
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

12. Triangle Kit — Sides & Angles

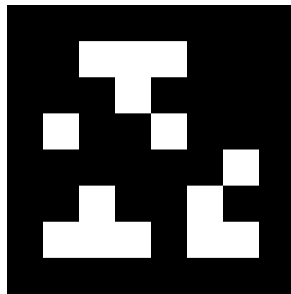
Classes 6–8

LEARNING OBJECTIVE

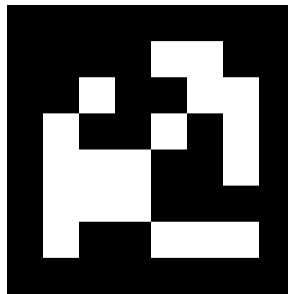
Measure and classify a physical triangle.



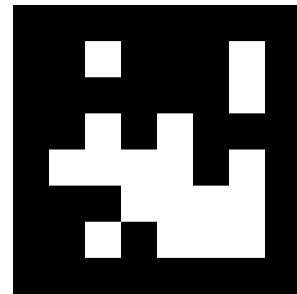
TAG #1
vertex A



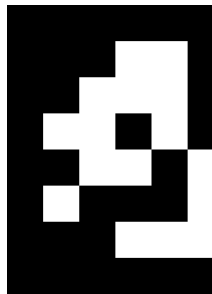
TAG #2
vertex B



TAG #3
vertex C



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

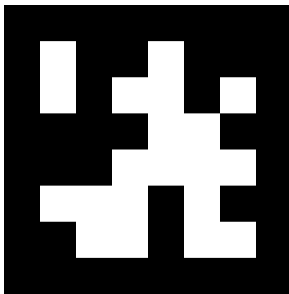
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

13. Integer Board — Addition & Subtraction

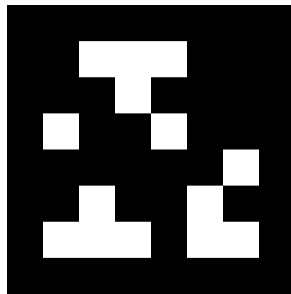
Classes 6–8

LEARNING OBJECTIVE

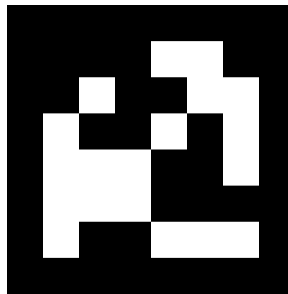
Locate positive and negative integers physically.



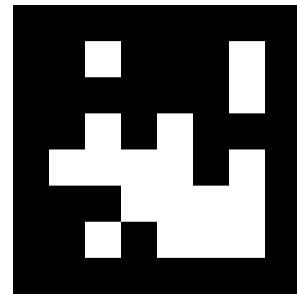
TAG #1
zero/reference



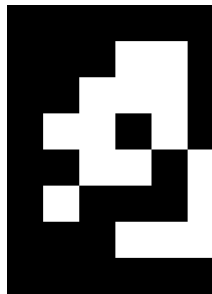
TAG #2
integer position



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

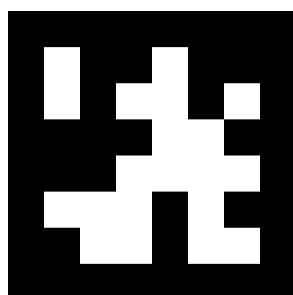
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

14. Unit Cubes — Volume

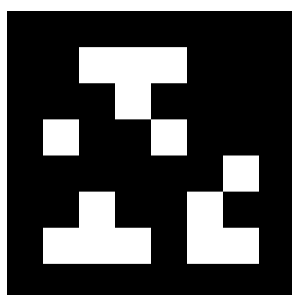
Classes 6–8

LEARNING OBJECTIVE

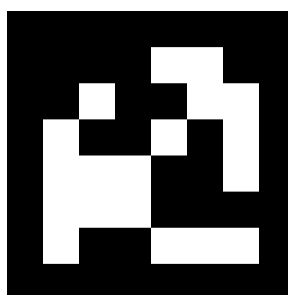
Model volume using physical unit cubes.



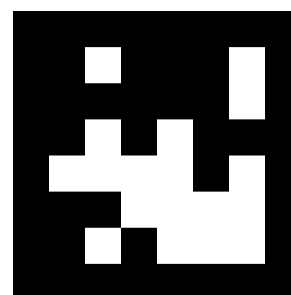
TAG #1
base



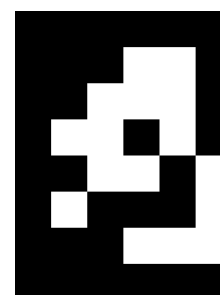
TAG #2
cube group



TAG #3
cube group



TAG #4
cube group



TAG #5
cube group

PHYSICAL TEST

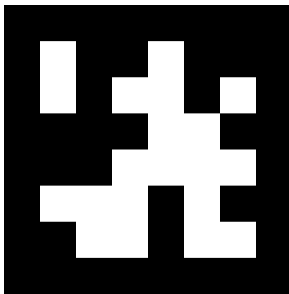
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

15. Geometry Box — Constructions

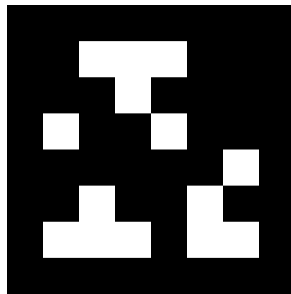
Classes 6–8

LEARNING OBJECTIVE

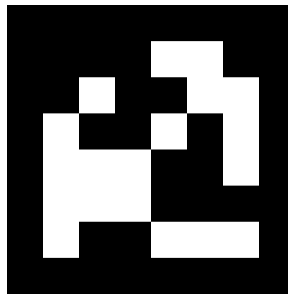
Use physical reference points to explore constructions.



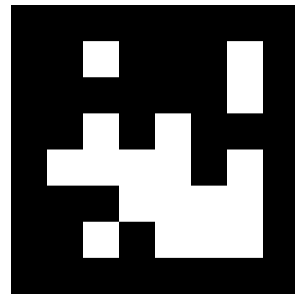
TAG #1
construction origin



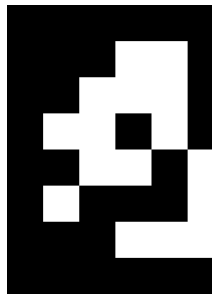
TAG #2
point A



TAG #3
point B



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

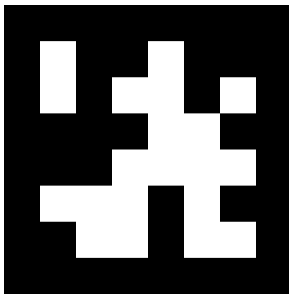
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

16. Geoboard & Rubber Bands — Area

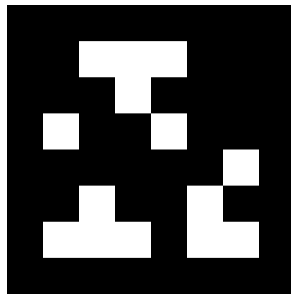
Classes 6–8

LEARNING OBJECTIVE

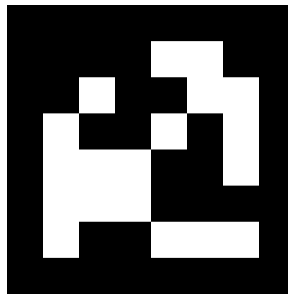
Create polygons and estimate area from physical vertices.



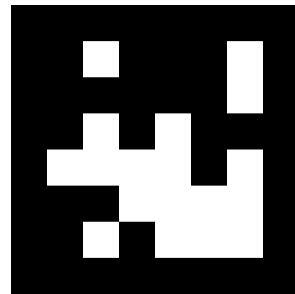
TAG #1
board origin



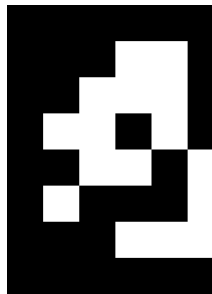
TAG #2
vertex



TAG #3
vertex



TAG #4
vertex



TAG #5
vertex

PHYSICAL TEST

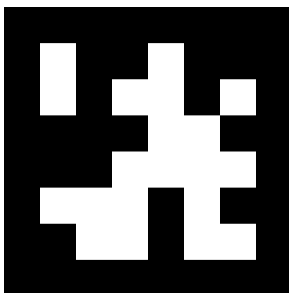
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

17. Graph Paper & Activity Materials

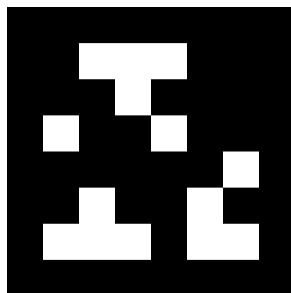
Classes 6–8

LEARNING OBJECTIVE

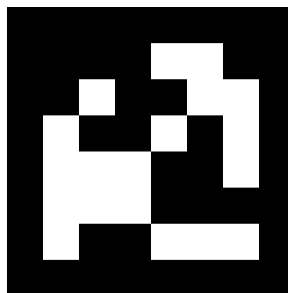
Map a physical point to graph coordinates.



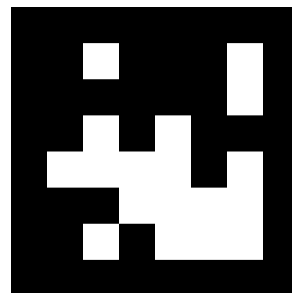
TAG #1
origin



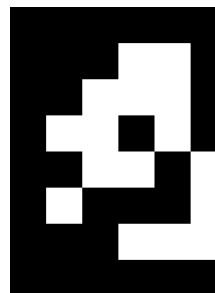
TAG #2
point



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

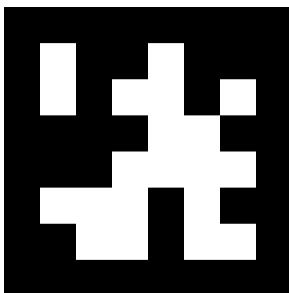
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

18. Probability Kit — Compound Events

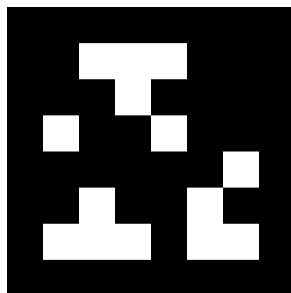
Classes 9–10

LEARNING OBJECTIVE

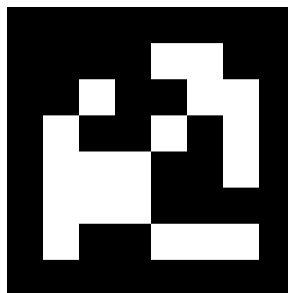
Model compound and experimental probability.



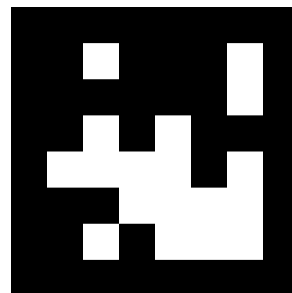
TAG #1
trial board



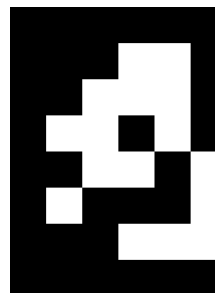
TAG #2
favourable



TAG #3
outcome



TAG #4
outcome



TAG #5
outcome

PHYSICAL TEST

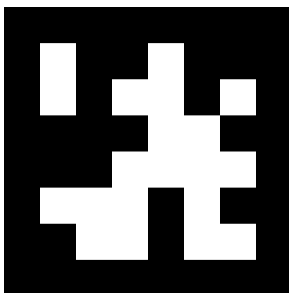
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

19. Mensuration Kit — Area & Volume

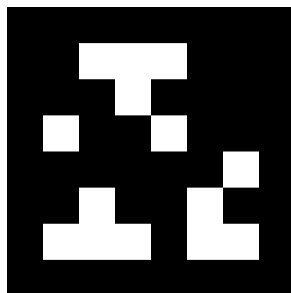
Classes 9–10

LEARNING OBJECTIVE

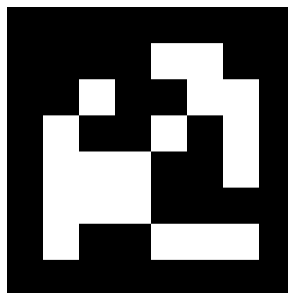
Estimate dimensions and calculate area or volume.



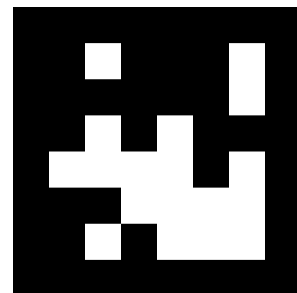
TAG #1
origin



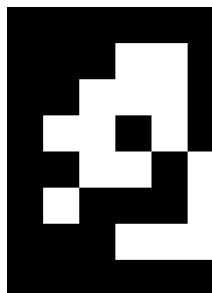
TAG #2
length



TAG #3
width



TAG #4
height



TAG #5
not used

PHYSICAL TEST

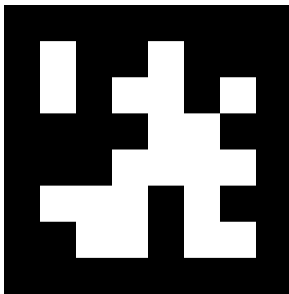
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

20. Measuring Jug Set — Capacity

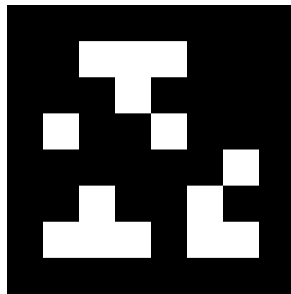
Classes 9–10

LEARNING OBJECTIVE

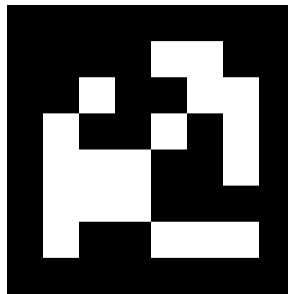
Explore capacity and measurement using a physical reference.



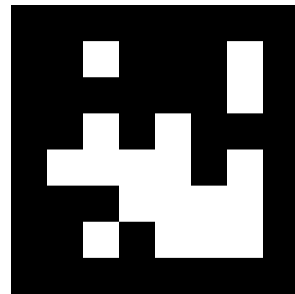
TAG #1
jug base



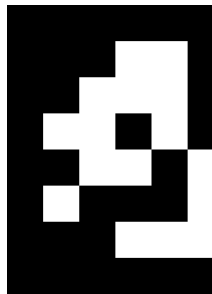
TAG #2
fill reference



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

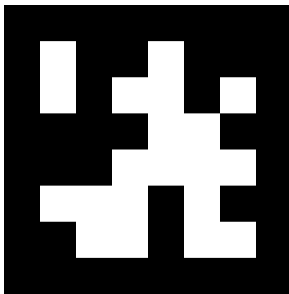
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

21. Geometry Box — Theorem & Construction

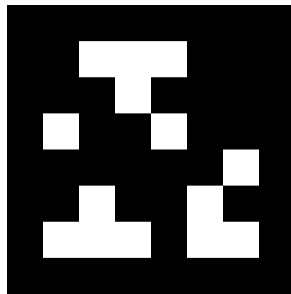
Classes 9–10

LEARNING OBJECTIVE

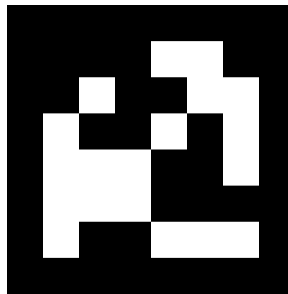
Connect physical construction steps with geometry.



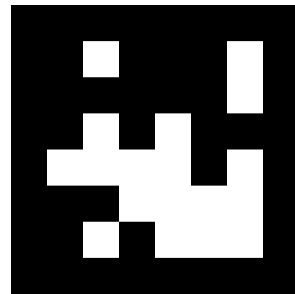
TAG #1
construction origin



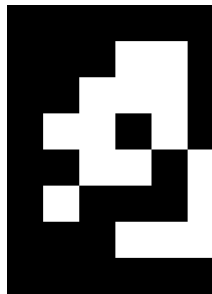
TAG #2
point A



TAG #3
point B



TAG #4
not used



TAG #5
not used

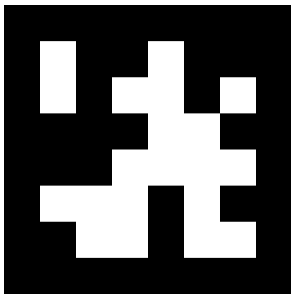
PHYSICAL TEST

- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

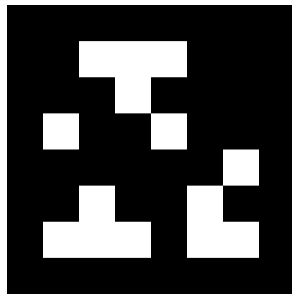
22. Mathematical Models / Charts — Formula Explorer Classes 9–10

LEARNING OBJECTIVE

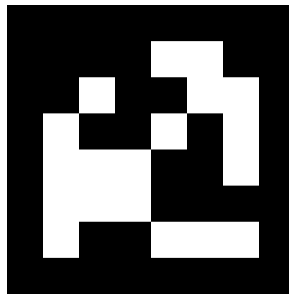
Explore formulas through physical model inputs.



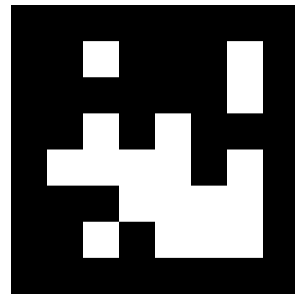
TAG #1
model base



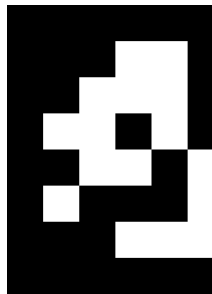
TAG #2
input A



TAG #3
input B



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

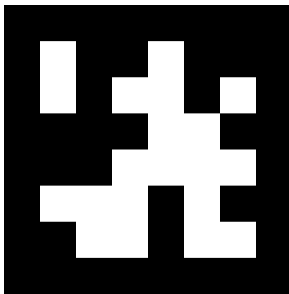
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

23. Graph & Isometric Sheets — 3D Coordinates

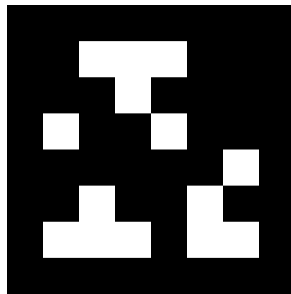
Classes 9–10

LEARNING OBJECTIVE

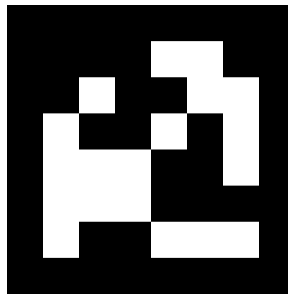
Map a tracked physical point into 3D coordinates.



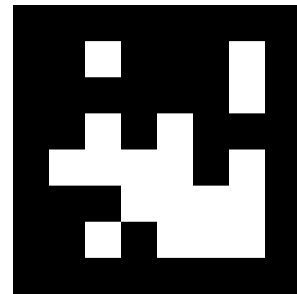
TAG #1
origin



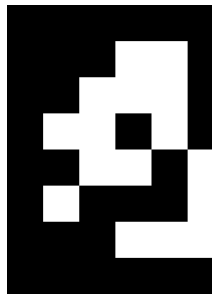
TAG #2
3D point



TAG #3
not used



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

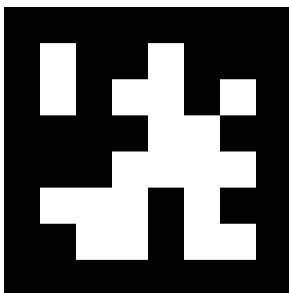
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

24. Dice, Coins & Playing Cards — Probability

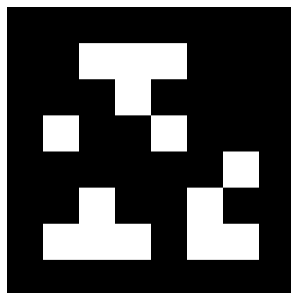
Classes 9–10

LEARNING OBJECTIVE

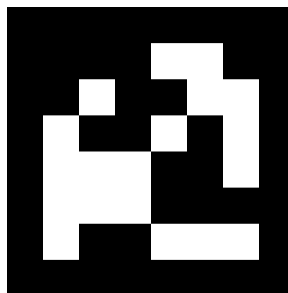
Use physical randomisation pieces to estimate probability.



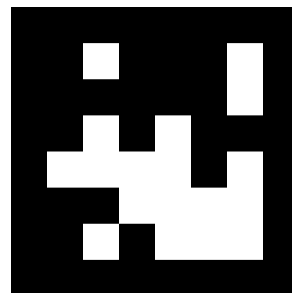
TAG #1
trial board



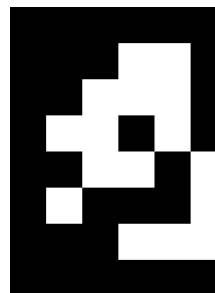
TAG #2
outcome A



TAG #3
outcome B



TAG #4
outcome C



TAG #5
outcome D

PHYSICAL TEST

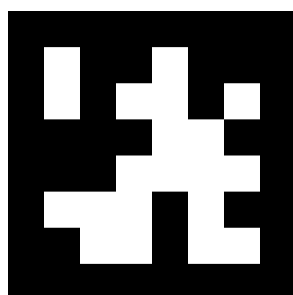
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

25. Conic Section Models — Circle, Ellipse, Parabola

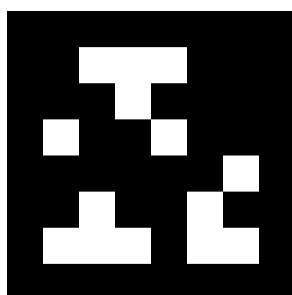
Classes 9–10

LEARNING OBJECTIVE

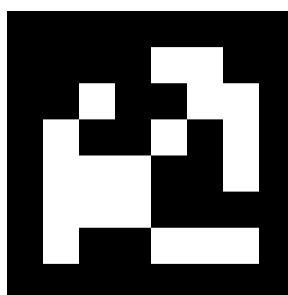
Compare the main conic-section forms.



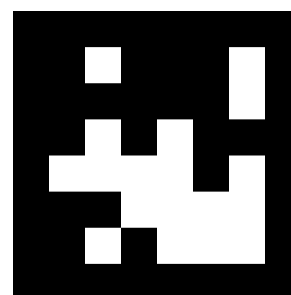
TAG #1
model base



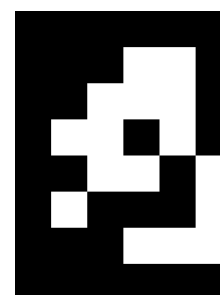
TAG #2
section marker



TAG #3
section marker



TAG #4
not used



TAG #5
not used

PHYSICAL TEST

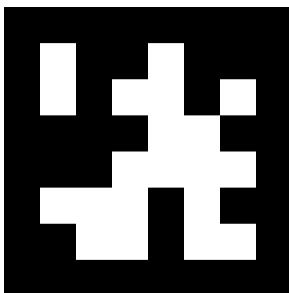
- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.

26. Venn Diagram Models / Charts — Sets

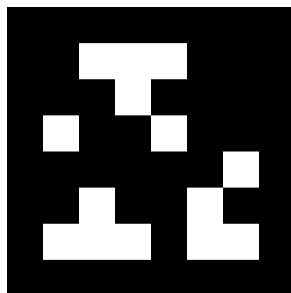
Classes 9–10

LEARNING OBJECTIVE

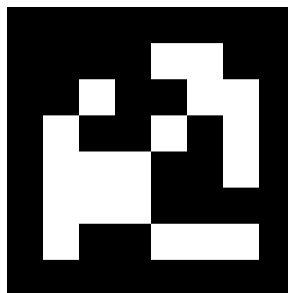
Classify physical set members and explore intersections.



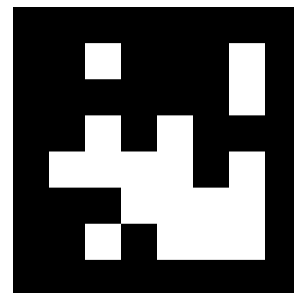
TAG #1
diagram base



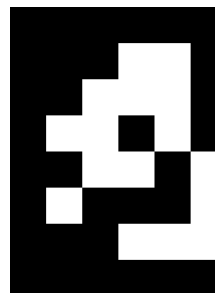
TAG #2
member



TAG #3
member



TAG #4
member



TAG #5
member

PHYSICAL TEST

- Print at 100% / Actual Size. Do not crop the black squares.
- Place each marker on the physical kit piece named above; keep Tag #1 as the base/reference when used.
- Point the USB webcam downward at the complete physical kit. The IFP should show the camera view with AR overlay.
- Expected state: LIVE PHYSICAL TRACKING when the required markers are visible.